SCHUR-CONVEXITY OF A RATIO OF MACDONALD POLYNOMIALS

SUVRIT SRA*

ABSTRACT. This note records a statement refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by exact arithmetic.

Summary: McSwiggen–Sahi Theorem 2.1: Macdonald lattice Schur-convexity (2026) *Certificate:* Rank-two Schur specialization, valid for every $0 < r < 1$.

Posed by McSwiggen and Sahi (2026); counterexample found by GPT-5.6 (Pro) (2026-05-14); formalized by S. Sra; audited by S. Sra.

Let $P_\lambda(x; q, t)$ denote the Macdonald polynomial in the monic normalization of Macdonald (1998), with parameters $q, t \in (0, 1)$ over a partition λ with at most n parts. At $q = t$ it specializes to the Schur polynomial $s_\lambda(x)$. The statement concerns two convexity properties of the map $\lambda \mapsto \Omega_\lambda(x; q, t)$ on partitions studied by McSwiggen and Sahi (2026): *log-convexity*, and *Schur-convexity* (i.e., $\lambda \succeq \mu$ implies $\Omega_\lambda \geq \Omega_\mu$). The normalization Ω_λ is defined inside the quoted statement.

Problem 1 (As posed: C. McSwiggen and S. Sahi (2026), Theorem 2.1)**.** For all $q, t \in (0, 1)$, $a > 0$, and $x \in \mathcal{L}_n^{q,t,a}$, the function $\lambda \mapsto \Omega_\lambda(x; q, t)$ is log-convex and Schur-convex.
Here, by §2.1 of the same paper, $\Omega_\lambda(x; q, t) = P_\lambda(x; q, t)/P_\lambda(t^\delta; q, t)$ is the Macdonald polynomial normalized to 1 at the principal specialization $t^\delta = (t^{n-1}, t^{n-2}, \ldots, 1)$, and the scaled, t-shifted, dominant q-lattice is

$$\mathcal{L}_n = \mathcal{L}_n^{q,t,a} = \left\{ (aq^{-\mu_1} t^{n-1}, aq^{-\mu_2} t^{n-2}, \ldots, aq^{-\mu_n}) : (\mu_1 \geq \ldots \geq \mu_n) \in \mathbb{Z}^n \right\} \subset \mathbb{R}^n.$$

Theorem 2 ([**Disproof**]: Theorem 2.1 of McSwiggen and Sahi (2026))**.** *The stated Schur-convexity fails already in rank two.*

Proof. Set $q = t = r \in (0, 1)$, $a = 1$, and choose lattice index $\nu = (1, 0)$. The resulting lattice point is $x = (1, 1)$. Let $\lambda = (2, 0)$ and $\mu = (1, 1)$; then $\lambda \succeq \mu$. At $q = t$, Macdonald polynomials specialize to Schur polynomials, and direct evaluation gives

$$\Omega_{(2,0)}(1, 1; r, r) = \frac{3}{1 + r + r^2}, \qquad \Omega_{(1,1)}(1, 1; r, r) = \frac{1}{r}.$$

But $1 + r + r^2 - 3r = (1 - r)^2 > 0$, so $\Omega_{(2,0)} < \Omega_{(1,1)}$ for every $0 < r < 1$, opposite to Schur-convexity. At $r = 1/2$, the values are $12/7 < 2$. □

Remark 3 (Scope). The specialization $q = t$ is a convenience. Appendix A carries the reversal for all $q, t \in (0, 1)$, with gap $(1 - t)^2(1 - qt)/(t(1 + t)(1 - qt^2))$, and shows it is admissible whenever $t = q^k$ for a positive integer k — the case above being $k = 1$. It also locates the failure, in a determinant shift that the principal-specialization normalization does not commute with, and shows that the

Date: 21 August 2026.

* TU Munich, Dept. of Mathematics, School of CIT; Garching, Germany. Email: s.sra@tum.de.

From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

1^n-normalized ratio

$$W_\lambda = P_\lambda(x)/P_\lambda(1^n)$$

satisfies in this case exactly the inequality Ω_λ violates.

Appendix A. Additional results for McSwiggen–Sahi Theorem 2.1

Theorem 2 refutes Schur-convexity at $q = t$, where Macdonald polynomials are Schur polynomials and the arithmetic is smallest. Nothing about the failure needs that specialization. This appendix records the reversal for all $q, t \in (0, 1)$, identifies the lattice points at which it is admissible, locates the step at which the claim goes wrong, and says what survives.

Throughout, $n = 2$, $\lambda = (2, 0)$ and $\mu = (1, 1)$, so that $\lambda \succeq \mu$, and the evaluation point is $x = (1, 1)$. Write $t^\delta = (t, 1)$.

The rank-two ratio. Only two Macdonald polynomials are involved, and neither is quoted: in degree two the triangularity $P_\lambda = m_\lambda +$ (lower terms) leaves a single unknown, and orthogonality in the (q, t) Hall inner product fixes it.

Lemma A.1. *For $n = 2$, $P_{(1,1)}(x_1, x_2; q, t) = x_1 x_2$ and*

$$P_{(2,0)}(x_1, x_2; q, t) = x_1^2 + x_2^2 + A\, x_1 x_2, \qquad A = \frac{(1+q)(1-t)}{1-qt}.$$

At $q = t$ this is $A = 1$, so $P_{(2,0)} = s_{(2)}$.

Proof. $P_{(1,1)} = e_2 = m_{(1,1)}$ is forced, being the lowest partition of 2. Write $P_{(2,0)} = m_{(2)} + A\, m_{(1,1)}$, and use coordinates in the power sums: $m_{(2)} = p_2$ and $m_{(1,1)} = e_2 = (p_1^2 - p_2)/2$. With $\langle p_\lambda, p_\mu\rangle_{q,t} = \delta_{\lambda\mu} z_\lambda \prod_i (1 - q^{\lambda_i})/(1 - t^{\lambda_i})$ we have $\langle p_2, p_2\rangle = 2(1 - q^2)/(1 - t^2)$, $\langle p_1^2, p_1^2\rangle = 2\big((1 - q)/(1 - t)\big)^2$ and $\langle p_2, p_1^2\rangle = 0$, so $\langle P_{(2,0)}, P_{(1,1)}\rangle = 0$ reads

$$A\,\frac{(1-q)^2}{(1-t)^2} = (2 - A)\,\frac{1-q^2}{1-t^2}, \qquad \text{i.e.} \qquad A\big[(1-q)(1+t) + (1+q)(1-t)\big] = 2(1+q)(1-t).$$

The bracket is $2(1 - qt)$, which gives A. □

Proposition A.2. *For all $q, t \in (0, 1)$,*

$$\Omega_{(2,0)}(1, 1; q, t) = \frac{2 + A}{1 + t^2 + At}, \qquad \Omega_{(1,1)}(1, 1; q, t) = \frac{1}{t},$$

and

$$\Omega_{(1,1)}(1, 1; q, t) - \Omega_{(2,0)}(1, 1; q, t) = \frac{(1-t)^2(1-qt)}{t\,(1+t)\,(1-qt^2)} \;>\; 0. \tag{A.1}$$

Proof. By Lemma A.1, $P_{(2,0)}(1, 1) = 2 + A$ and $P_{(2,0)}(t, 1) = 1 + t^2 + At$, while $P_{(1,1)}(1, 1) = 1$ and $P_{(1,1)}(t, 1) = t$, which gives the two values. Subtracting,

$$\frac{1}{t} - \frac{2 + A}{1 + t^2 + At} = \frac{1 + t^2 + At - t(2 + A)}{t(1 + t^2 + At)} = \frac{(1 - t)^2}{t(1 + t^2 + At)},$$

the term At canceling — so the gap does not depend on A except through the denominator. Substituting A and clearing,

$$1 + t^2 + At = \frac{(1 + t)(1 - qt^2)}{1 - qt},$$

which turns the previous display into (A.1). On $q, t \in (0, 1)$ each of $(1 - t)^2$, $1 - qt$, t, $1 + t$ and $1 - qt^2$ is strictly positive. □

At $q = t = r$ this is $\frac{1}{r} - \frac{3}{1+r+r^2} = \frac{(1-r)^2}{r(1+r+r^2)}$, the computation carried in Theorem 2.

Where the point is admissible. The reversal above is an identity in q and t, but it refutes the claim only where $x = (1, 1)$ is a legal evaluation point, i.e. lies on $\mathcal{L}_n$.

Lemma A.3. *Let $t = q^k$ with k a positive integer and $a = 1$. Then $1^n \in \mathcal{L}_n^{q,q^k,1}$.*

Proof. Take $\mu_i = k(n - i)$, which is a partition: $\mu_i - \mu_{i+1} = k > 0$. The i-th coordinate of the corresponding lattice point is $q^{-\mu_i} t^{n-i} = q^{-k(n-i)} q^{k(n-i)} = 1$. □

So for every positive integer k the pair $\lambda = (2, 0) \succeq \mu = (1, 1)$ at $x = (1, 1)$ refutes Schur-convexity for the parameters (q, q^k), and Theorem 2 is the case $k = 1$. For an instance away from the Schur locus take $k = 2$, $q = 1/2$, $t = 1/4$: then $A = 9/7$ and

$$\Omega_{(2,0)}(1, 1; \tfrac{1}{2}, \tfrac{1}{4}) = \frac{368}{155} < 4 = \Omega_{(1,1)}(1, 1; \tfrac{1}{2}, \tfrac{1}{4}),$$

a gap of $252/155$.

Where the claim goes wrong. The failure is not an accident of small rank: it is the principal-specialization normalization, which does not commute with the determinant shift.

Lemma A.4 (Determinant shift for Ω)**.** *Let $\lambda = (\lambda_1, \ldots, \lambda_N)$ and $c = \lambda_N$. Then*

$$\Omega_\lambda^{(N)}(y; q, t) = \left(\frac{y_1 \cdots y_N}{t^{N(N-1)/2}} \right)^c \Omega_{\lambda - c(1^N)}^{(N)}(y; q, t).$$

Proof. From $P_\lambda(y) = (y_1 \cdots y_N)^c P_{\lambda - c(1^N)}(y)$, evaluation at t^δ contributes $\left(\prod_{i=1}^N t^{N-i}\right)^c = t^{cN(N-1)/2}$, since $\sum_{i=1}^N (N - i) = N(N - 1)/2$. Divide the two relations. □

The factor $t^{-cN(N-1)/2}$ is exactly what the argument for Schur-convexity drops. Its effect is not a perturbation. At $N = 2$ and $\lambda = (1, 1)$, where $c = 1$ and $\lambda - c(1^2) = (0, 0)$, the lemma gives $\Omega_{(1,1)}(y) = y_1 y_2/t$, whereas the shift without the t-power would give $y_1 y_2$; the two agree only at $t = 1$. Evaluated at $y = 1^2$ this is the $1/t$ of Proposition A.2, which diverges as $t \downarrow 0$ while $\Omega_{(2,0)}(1, 1; q, t) \to 3 + q$ stays bounded. The reversal is therefore not delicate: past a point the two sides are not even the same order of magnitude.

What is not refuted. Normalizing at 1^n instead of at t^δ removes the phenomenon entirely. Write $W_\lambda(x; q, t) = P_\lambda(x; q, t)/P_\lambda(1^n; q, t)$. Since $P_\lambda(1^N) = P_{\lambda - c(1^N)}(1^N)$, the determinant shift for W carries no extra factor at all, and $W_\lambda(1^n; q, t) = 1$ for every λ — so the witness above says nothing whatever about W, both sides being 1 at the evaluation point. More than that, in the same rank-two case the W-inequality holds where the Ω-inequality fails:

$$W_{(2,0)}(x_1, x_2; q, t) - W_{(1,1)}(x_1, x_2; q, t) = \frac{x_1^2 + x_2^2 + A x_1 x_2 - (2 + A) x_1 x_2}{2 + A} = \frac{(x_1 - x_2)^2}{2 + A} \;\geq\; 0.$$

This is the rank-two shadow of a general fact on the Schur locus $q = t$, where W_λ is the normalized Schur function $s_\lambda(x)/s_\lambda(1^n)$: for every x with nonnegative entries, $\lambda \succeq \mu$ implies $s_\lambda(x)/s_\lambda(1^n) \geq s_\mu(x)/s_\mu(1^n)$, which is the conjecture of Cuttler, Greene and Skandera, proved by Sra (2016). So on that locus the 1^n-normalized ratio is Schur-convex in exactly the sense that Ω_λ is not. Theorem 2 is thus a statement about the normalization, not about Macdonald polynomials on the lattice: what it refutes is Schur-convexity of $\lambda \mapsto \Omega_\lambda$, and any corresponding claim for W_λ is untouched.

The certificate `counterexamples/macdonald-schur-convexity/verify.py` recomputes all of this exactly: it recovers A from the (q, t) Hall inner product rather than quoting Lemma A.1, verifies (A.1) as a rational-function identity in q and t together with the factorization of $1 + t^2 + At$ that makes each sign visible, checks the lattice membership of Lemma A.3 symbolically in q, k and n, checks the shift exponent and the $N = 2$ instance where the naive shift fails, and confirms the W identity above.

Code and certificate

The case is the directory `counterexamples/macdonald-schur-convexity/` of the archive, at commit `7b17a26`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. Independent Sage interval cross-checks: `verify_macdonald.sage`. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/macdonald-schur-convexity
python verify.py
```

- Theorem 2 (result `macdonald-schur-convexity`): exact arithmetic.
- `artifacts/certificate.json`: Exact rank-two reversal at sample parameters, the general (q,t) gap identity, lattice membership, the determinant-shift diagnosis, and the 1^n-normalized scope check.

AI usage

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexample here was found by GPT-5.6 (Pro) (2026-05-14). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

Acknowledgements

The author (SS) thanks the Alexander von Humboldt (AvH) Foundation for their generous support via a Humboldt Professorship in AI.

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